

# Thuan Hieu Tran

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## EDUCATION

### University of Chicago

B.S. in Computer Science & Economics | Full-Ride Award

Chicago, IL

Expected June 2030

### Hanoi–Amsterdam High School for the Gifted

Hanoi, Vietnam

## HONORS & AWARDS

**Silver Medal**, International Artificial Intelligence Olympiad (Ljubljana, Slovenia; UNESCO/AWS)

**Third Prize**, ICPC Asia Regional; **First Prize (7th of 200 teams)**, ICPC Vietnam (High School Division)

**National Champion (1st of 2,200)**, Vietnam National Young Informatics Contest

**Champion**, Vietnam AI Championship; **Second Prize**, Vietnam Olympiad in AI

**National Third Prize (top 8%)**, Vietnam Olympiad in Informatics (VOI); **Codeforces Master** (rating 2104)

## EXPERIENCE

### Software Engineer & Project Lead

Feb. – Aug. 2026

*IterLight*

Florida, USA (Remote)

- Revived a previously failed initiative as project lead: built a custom harness for a human-in-the-loop AI pipeline that auto-generates TikTok videos from the company's proven viral format
- Scaled the pipeline from 1 to 5 TikTok channels with a scheduled multi-account posting system; the top channel passed 3M views at roughly 90% lower cost than building an in-house production team
- Solo-built a RAG sales and support pipeline for Facebook Messenger: architected a 138-page linked-wiki knowledge base, designed a human-review workflow and unlocked personal-account chats beyond the official API via Matrix
- Ran continuous pilot-and-refine cycles with the sales team, doubling reply speed and raising the conversion rate 1.5x while replacing outsourced sales agents at 60% under projected API token cost

### Research Assistant

2025

*AI4LIFE Institute, Hanoi University of Science and Technology*

Hanoi, Vietnam

- Built the data preprocessing pipeline for a spatial gene expression model on H&E histology slides, turning 36 whole-slide images from 8 patients into 9,612 spot-level patches across 785 genes
- Ran LoRA fine-tuning experiments on the UNI pathology foundation model; final model reached 3.5x baseline PCC

### Algorithm Optimization Intern

June – Sep. 2024

*AimeSoft*

Hanoi, Vietnam

- Optimized timetables for a Japanese rail operator (outsourced) via greedy-initialized simulated annealing
- Reformulated the objective function to incorporate previously unused sensor data; validated schedules in a discrete-event simulation, cutting worst-case passenger wait time by up to 17%

## RESEARCH & PUBLICATIONS

### Sign Language Recognition: A Large-scale Multi-view Dataset and Comprehensive Evaluation

*Co-Author*

WACV 2025

- Co-developed Multi-VSL, a pioneering multi-view Vietnamese Sign Language dataset (84K+ videos, 1,000 glosses, 30 signers) that became a foundation for later VSL research and products in Vietnam; 26 citations
- Set up the 3-camera recording studio and built the annotation platform; cross-tested I3D, Swin, MVITv2 and VTN-PF across views, showing multi-view fusion lifts top-1 accuracy by up to 19.75 pp

## PROJECTS

### LittleHands | Co-Creator & Lead Developer

June 2024 – Present

*A smartphone app that translates Vietnamese Sign Language to and from text for deaf and hard-of-hearing users*

- Trained Video Transformer Networks on our Multi-VSL dataset for sentence-level sign recognition
- Built the FastAPI + PostgreSQL backend and C# client for two-way sign, text and animation translation

### Bottleneck-Migration Alpha

July 2026

- Built a market-neutral long/short sector-rotation strategy on the AI hardware supply chain, with price-based regime detection, beta hedging and volatility targeting; backtested Sharpe 1.30 net of costs

## TECHNICAL SKILLS

**Languages & ML:** Python, C/C++, Java, PyTorch, CUDA, NumPy, pandas, Hugging Face (PEFT/LoRA), Unsloth

**Infrastructure:** Google Cloud, Docker, Matrix (Synapse), Linux/Bash, Git